

Let's Encrypt HTTPS — Integration Status

Revision: 2 **Last modified:** 2026-07-01T10:06:00Z **Status:** In progress — Phases 0–3 PASS: a REAL hermetic DNS-01 certificate is issued by local Pebble and verified by the cert-analyzer (re-runnable). Phase 5 (renewal/rotation) next; Phase 4 (LE staging) + Phase 6 (production cutover) OPERATOR-BLOCKED (design §9 real-domain gate). **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc for the Let's Encrypt HTTPS workstream (task #59). **Companion:** design ../LETSENCRYPT_HTTPS.md · plan ../LETSENCRYPT_HTTPS_PLAN.md · summary Status_Summary.md

Operator-blocked items (read first — §11.4.45 O(1) surface)

Item	Why blocked	Unblock condition
Phase 4 — LE staging end-to-end (real DNS-01)	Needs a real DNS provider + an API token stored as a Podman secret (§11.4.10 — value never in git/.env). Operator chose <i>defer real domain</i> (Phase 0).	Operator provides the DNS provider name + a scoped API token, and names a test hostname.
Phase 6 — production cutover	Real public domain, real DNS-01 for that domain, single operator-gated go-live (design §9).	Operator authorises the go-live with the real domain + token + host firewall opening.

Operator decisions captured (Phase 0 — §11.4.66)

Decision	Choice	Date
		2026-07-01

Decision	Choice	Date
TLS client / terminator	Caddy auto-HTTPS (in-process ACME client + terminator; no cron/CI/hook renewal — §11.4.156)	
ACME challenge type	DNS-01 (works without inbound :80/:443 to the host)	2026-07-01
Rollout order	Hermetic (Pebble) + LE-staging first; defer real domain — prod is one later operator-gated cutover	2026-07-01

Phase status (captured-evidence-driven — §11.4.5/§11.4.69)

Phase	Scope	Validation	Evidence
0 — Operator decisions + scaffolding	Capture the 3 design §9 decisions; scaffold the plan	PASS	decisions table above; committed LETSENCRYPT_HTTPS.md + _PLAN.md @ fddd25c
1 — Cert-analyzer (client/challenge-agnostic)	Pure offline analyzer over a resulting PEM (cert_not_expired / cert_days_remaining / cert_san_matches / cert_chain_roots_in / cert_renewal_due) + golden-good/golden-bad self-validation (§11.4.107(10))	PASS	selftest 37/37; guard GREEN+RED PASS; §1.1 mutation (SAN exact→substring) → guard FAIL, restore byte-identical md5 a06e0f89...; qa-results/letsencrypt/cert-analyzer/qa-results/regression/cert_analyzer_selfvalidation
1 — Custom Caddy image w/ DNS-01 provider module	Build a rootless Caddy image bundling the local dns.providers.challtestsrv module (§11.4.161)	PASS	build.sh built localhost/helix_proxy/caddy-challtestsrv:2.8.4; post-build anti-bluff check caddy list-modules grep dns.providers.challtestsrv PASS (a build ≠ module linked, §11.4.6); qa-results/letsencrypt/build/
2 — Compose service + Podman secret + cert volume +	Wire Caddy + CoreDNS (zone-determination fix) + secret NAME/PATH (no value) + persistent cert volume	PASS	compose.hermetic.yml (4 services; PEBBLE_VA_ALWAYS_VALID=0; secrets commented for hermetic; caddy loopback bound per security-audit M1); CoreDNS answers hermetic.test SOA, forward

Phase	Scope	Validation	Evidence
CoreDNS SOA front			_acme-challenge TXT to challtestsrv re-runnable phase3_hermetic_issue.sh → clean runs each issued a n cert (Pebble VA_ALWAYS_VALID=0 genuine validation); cert- analyzer PASS: not_expire + SAN proxy.hermetic.test negative-SAN-reject + chain-to-this-run's- Pebble-CA ; qa-results/ letsencrypt/phase3_issuanc 20260701T100119Z/ + ... T100434Z/
3 — Hermetic local- ACME with Pebble (anti-bluff core)	Caddy issues a REAL cert against local Pebble via DNS-01, fully offline; cert-analyzer asserts the served leaf	PASS	
4 — LE staging end-to-end (real DNS-01)	Real LE staging endpoint + real DNS-01	OPERATOR- BLOCKED	needs DNS provider + tok (\$11.4.10)
5 — Renewal + rotation simulation (zero- downtime)	Force near-expiry → observe Caddy renew → analyzer confirms new cert; no downtime	PENDING	—
6 — Production cutover + rollback runbook	Real domain go-live	OPERATOR- BLOCKED	operator go-live gate (des \$9)

Honest boundary (§11.4.6)

Phases 0–3 are DONE with captured evidence: the custom Caddy image is built with the DNS-01 module linked (anti-bluff-verified), the hermetic stack is wired (with the CoreDNS SOA front that fixes certmagic's zone-determination), and phase3_hermetic_issue.sh issues a REAL cert via DNS-01 against local Pebble (genuine validation — PEBBLE_VA_ALWAYS_VALID=0) that the cert-analyzer verifies over the served leaf, including that it chains to THIS RUN's Pebble CA — proven on two clean re-runnable runs (§11.4.98 / §11.4.107). What is NOT yet exercised (honestly): Phase 5 renewal/rotation (research landed, docs/research/letsencrypt_renewal_20260701/; the running stack is left up

via KEEP_UP=1 for it) and the real-domain paths (Phase 4 LE staging, Phase 6 production cutover) which wait on the operator gate — never a metadata-only PASS (§11.4 / §11.4.1).